

Exam Date & Time: 20-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Pattern Recognition [CSE-4036]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) How would you apply the principles of pattern recognition to develop a system capable of identifying handwritten digits? (5)
 - A)
 - B) Analyze the role of sensory data in machine perception. How does sensory data influence the accuracy and effectiveness of applications such as facial recognition and autonomous driving? (3)
 - C) Examine how classification and clustering techniques address different types of data analysis tasks. Why is each method important in machine learning? (2)
- 2) *Develop a Bayesian classification model to diagnose tomato leaf diseases, where the classes are C_1 : Early Blight, C_2 : Late Blight, and C_3 : Septoria Leaf Spot.*
 - A) *Given the following information:*
 - Prior probabilities: $P(C_1) = 0.5$, $P(C_2) = 0.3$, and $P(C_3) = 0.2$.
 - Likelihood probabilities based on observed symptoms X : $P(X | C_1) = 0.6$, $P(X | C_2) = 0.25$, $P(X | C_3) = 0.15$.*Loss matrix:*
 - Misclassifying Early Blight as Late Blight or Septoria costs 4 units.
 - Misclassifying Late Blight costs 3 units.
 - Misclassifying Septoria Leaf Spot costs 1 unit.*Using this information, calculate the expected risk for each decision rule, synthesize an optimal decision rule that minimizes expected risk, and justify the classification decision in terms of effective disease management.* (5)
 - B) (3)

Suppose the daily study hours of a random sample of 15 college students are recorded as follows (in hours):

$$X = \{3, 4, 5, 2, 6, 4, 5, 7, 3, 5, 4, 6, 2, 3, 5\}$$

Determine the Maximum Likelihood Estimators (MLE) for the mean μ and the standard deviation σ of the population, assuming the study hours are normally distributed. Explain each step involved in calculating the MLEs and interpret the results.

- C) Describe the difference between univariate and multivariate density functions. How does each type represent data distribution? (2)

- 3) Given two classes of data points, C_1 and C_2 , with the following feature vectors:

A) Class C_1 :

$$X_1 = \{(1, 2), (2, 3), (3, 3), (2, 2), (3, 4)\}$$

Class C_2 :

$$X_2 = \{(6, 5), (7, 8), (8, 7), (7, 6), (8, 8)\}$$

Apply Fisher's Linear Discriminant Analysis (LDA) to find the optimal projection direction w that maximizes the separation between the two classes. Calculate the within-class scatter matrix S_W and the between-class scatter matrix S_B . Determine the projection vector w by using these matrices and interpret the resulting direction in terms of its effectiveness for class separation. (5)

- B) Consider a fuzzy classification problem where we are classifying temperatures into three categories: **Cold**, **Warm**, and **Hot**. The temperature values (in degrees Celsius) and their fuzzy membership values for each category are given below:

(i) Temperature = 15°C : Cold = 0.8, Warm = 0.3, Hot = 0.0

(ii) Temperature = 25°C : Cold = 0.2, Warm = 0.7, Hot = 0.1

(iii) Temperature = 35°C : Cold = 0.0, Warm = 0.4, Hot = 0.9

(3)

Analyze the fuzzy membership values for each temperature to determine the most appropriate classification for each temperature based on the highest membership value

- C) Explain the purpose of Parzen windows in density estimation. How does the choice of window size affect the smoothness of the estimated probability density function? (2)

- 4)
- A) Analyze a Hidden Markov Model (HMM) with three states S_1 , S_2 , and S_3 , and two possible observations O_1 and O_2 . The initial state probabilities are $\pi = [0.6, 0.3, 0.1]$, the state transition matrix A is:
- $$A = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.4 & 0.5 & 0.1 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}$$
- and the emission probability matrix B is:
- $$B = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \\ 0.5 & 0.5 \end{bmatrix} \quad (5)$$
- Given an observation sequence $O = [O_1, O_2, O_1]$, apply the forward algorithm to compute the probability of this observation sequence. Interpret the result in terms of how likely this sequence is to occur given the HMM parameters.
- B) Describe the steps involved in calculating the Mean Squared Error (MSE) for a dataset with predicted and actual values. What information does MSE provide about the model's performance? (3)
- C) Compare the roles of the prior and likelihood in Bayesian Estimation. How do they influence the posterior distribution? (2)

- 5)
- A) Apply the K-means clustering algorithm to the following dataset with five points in a 2D space:
- $$X = \{(2,3), (3,4), (5,6), (8,8), (9,7)\}$$
- Set $k = 2$ (two clusters) and the initial centroids as $(2,3)$ and $(9,7)$. Describe each step in the K-means algorithm, including the assignment of points to clusters and the calculation of new centroids. Determine the final clusters. (5)
- B) A model is trained to classify images of animals into three classes: **Cat**, **Dog**, and **Rabbit**. After testing the model on a set of 100 images, the following results are observed:
- 30 images are of **Cats**, 35 are of **Dogs**, and 35 are of **Rabbits**.
 - The model's predictions for these 100 images are summarized in the following confusion matrix: (3)

		Predicted		
		Cat	Dog	Rabbit
Actual	Cat	20	5	5
	Dog	5	25	6
	Rabbit	6	3	26

Analyze the confusion matrix and give your observations.

- C) *Explain* the purpose of clustering in data analysis. How does clustering *group* data points based on similarity? (2)

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